

GT-Code

A coding agent that runs entirely on the consultant's machine, on local open-weight models. No cloud LLM, no API key, no telemetry.

The useful version of a coding agent inside a firm like ours is the one you can point at a client folder. That rules out sending the folder to a cloud model, so the constraint came first and the architecture is what fell out of it.

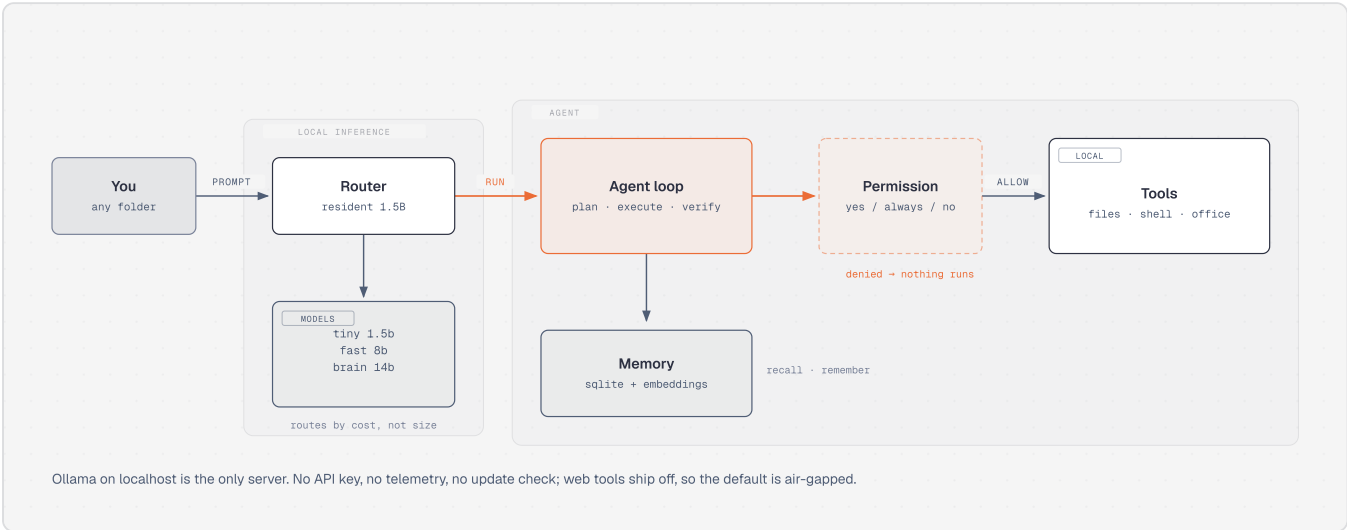


Fig. 1 – end to end. Ollama on localhost is the only server.

Three decisions

DECISION	WHY
A ladder, not one big model 1.5b → 8b → 14b	Every model swap costs 10–60 s of loading before the first token, which is what makes local agents feel unusable. A resident 1.5B routes and handles everyday work; the 8B escalates; the 14B is reserved for architecture and planning. Nothing above 14B — on this hardware it is not an interactive agent.
Apache-2.0 throughout Qwen + nomic-embed	No Meta Llama licence anywhere in the line-up, so nothing needs a licence review before it goes on a firm laptop.
The permission gate is code	Standing grants persist on disk and are coarse enough to read. A destructive-command regex overrides all of them and always prompts, even under auto-approve. Hooks go further: a <code>pre_tool</code> hook that exits 2 blocks the call outright.
A confidence gate before a build	One small model call weighs the plausible readings of a request and scores its confidence. High builds now, medium plans and waits, low asks one question. Never fires mid-task, and fails open — the gate can only add care.

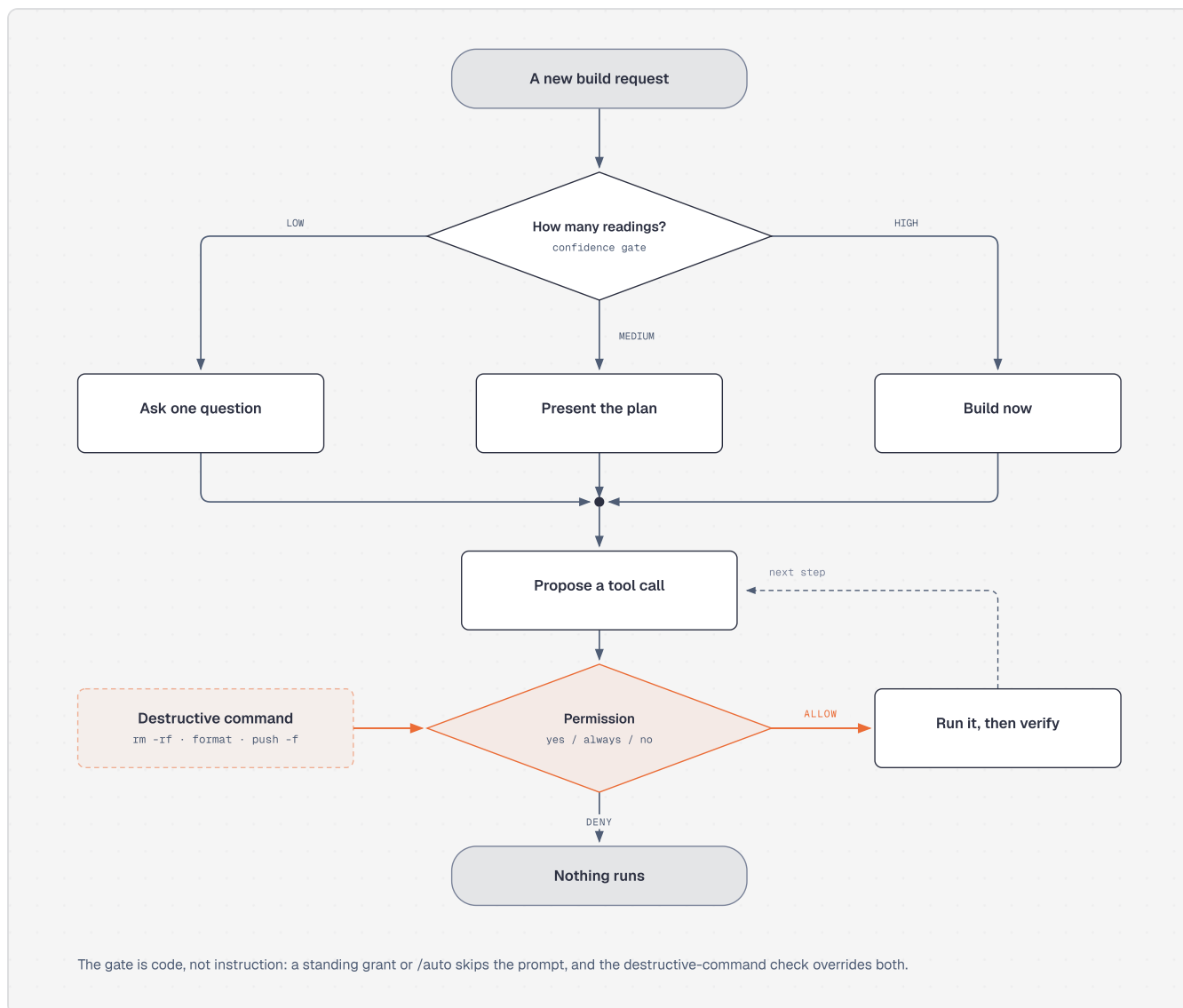


Fig. 2 – the two places the agent is allowed to stop.

The rest of it

PIECE	WHAT IT DOES
Agent loop	Native function calling where the model's template supports it, a portable JSON protocol everywhere else.
Memory	nomic-embed vectors in SQLite, brute-force cosine in numpy — nothing needing a native build on Windows.
Project memory	A GT.md per project, loaded every turn.
Compaction	Old turns distilled into a rolling summary rather than dropped, so the goal from the start of a long build survives.
Sub-agents	Read-only research in a separate context; only the report returns. Isolation, not parallelism.
Skills	Eight curated playbooks injected on keyword match. Two sharp pages beat gigabytes of retrieved docs in an 8K window.

Push on this

- The confidence gate is one model call scoring itself. No measurement of how well that score correlates with actually-misread requests — only that it fails open.
- The hybrid tool protocol is two code paths for one behaviour. Only one is exercised on any given model.
- 480+ dependency-free logic tests, and none that run a real model. The model-dependent behaviour is the part I cannot regress on.

The local-first constraint is what makes this deployable, and it is also what caps its capability. I would rather argue about that trade than hide it.